

$$\text{Irreducible } f(x) \in \mathbb{F}_p[x]$$

(Proposition from Lecture 9)

$$\deg f(x) = r \mid d$$

$$|k| = p^d$$

Then $f(x)$ has r distinct zeros in k .

Example:

$$x^3 - 2 \in \mathbb{Q}[x] \text{ is irreducible}$$

$$\alpha \in k = \frac{\mathbb{Q}[x]}{(x^3 - 2)} \ni \alpha = x + (x^3 - 2)$$

$$\alpha^3 - 2 = x^3 - 2 + (x^3 - 2) = 0$$

is a zero for

$$x^3 - 2$$

$$\mathbb{Q}[x]$$

$\downarrow$

$$k = \frac{\mathbb{Q}[x]}{(x^3 - 2)}$$

$$\longrightarrow \mathbb{R}$$

$$e\sqrt[3]{2} : x = \sqrt[3]{2} \text{ in fact}$$

$\sqrt[3]{2}$ is the only zero for

$$\alpha \mapsto \sqrt[3]{2} \quad x^3 - 2 \text{ in } \mathbb{R}$$

There are two other zeros
for $x^3 - 2$ in $\mathbb{C}$

$$\cdot \alpha \mapsto e^{2\pi i/3} \sqrt[3]{2}$$

$$\cdot \alpha \mapsto e^{4\pi i/3} \sqrt[3]{2} \in \mathbb{C}$$

Recall of proposition

$$f(x) \mid x^{pd} - x$$

pd distinct
zeros in k

Every elt
 $\alpha \in k$
satisfies $\alpha^{pd} = \alpha$
 $\Rightarrow$

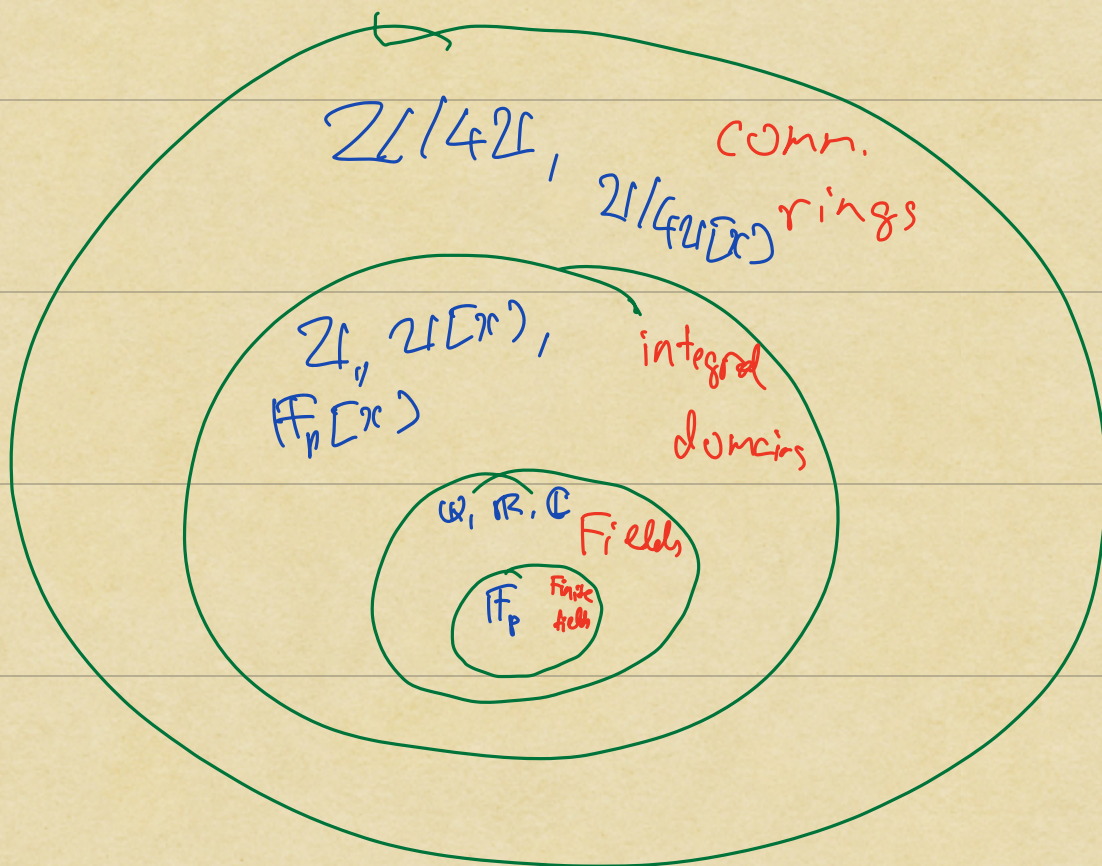
$$x^{pd} - x = f(x) g(x) \in \mathbb{F}_p[x]$$

pd distinct
zeros

$\leq r$
zeros
in k

$\leq pd - r$
zeros
in k

Summary



• Ideals:

R : comm. ring, $I \leq R$: subgroup

Defⁿ I is an ideal if
 $\forall r \in R, j \in I, rj \in I$.

TF AE: (1) I is an ideal
 (2) $R \twoheadrightarrow R/I$ is a ring homomorphism
 (3) $I = \ker \varphi$, for some $\varphi: R \rightarrow R'$ ring hom.

• Polynomial rings

(a) $R[x]$, (R : comm. ring)

$$\left\{ \begin{array}{l} \text{Ring hom.} \\ R[x] \rightarrow R' \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \lambda: R \rightarrow R' \\ \underbrace{r \in R'}_{\uparrow} \end{array} \right\}$$

$\in V_{\lambda, r}: \sum a_i x^i \mapsto \sum \lambda(a_i) r^i$

(b) K : field

$K[x]$ has the following properties:

(i) Every ideal is **principal**
i.e. is of the form

$$(f(x)) = \{ a(x)f(x) : a(x) \in K[x] \}$$

(ii) TRUE: (1) $f(x)$ is irreducible
[deg $f(x) \geq 1$] (2) $\underline{K[x]}$ is a field

(3) $\frac{k[x]}{(f(x))}$ is an integral domain

(iii) (Unique factorization)

$$0 \neq g(x) \in k[x]$$

$$g(x) = q_1(x) q_2(x) \dots q_r(x) \\ \in k[x]$$

with $q_i(x)$ irreducible $\nexists c$

& these irreducible factors

are unique up to permutations
& constants

(iv) (Division & Bezout)

monic $0 \neq f(x) \in k[x], g(x) \in k[x]$

$$g(x) = q(x)f(x) + r(x), \\ \deg r(x) < \deg$$

$$4 \quad \gcd(f(x), g(x)) = s(x)f(x) + t(x)g(x)$$

Finite fields

k : finite field

(1) $\mathbb{F}_p \subseteq k$, for some prime p

(2) $|k| = p^d$, $\exists d \geq 1$

(3) $k^\times$ is cyclic $\Leftarrow \left(\begin{array}{l} x^n - 1 \\ \text{has at} \\ \text{most} \\ n \text{ zeros in } k \\ \text{FT of} \\ \text{f.g. abelian} \\ \text{groups} \end{array} \right)$

(4)

$\exists$ irred. polynomial

$$f(x) \in \mathbb{F}_p[x] \text{ i.t.}$$

$$\frac{\mathbb{F}_p[x]}{(f(x))} \xrightarrow{\cong} K.$$

$\deg d$

(5)

$$\text{Fr}: K \longrightarrow K$$

$\alpha \longmapsto \alpha^q$

Frobenius automorphism

(6)

Galois' thm

$\exists$ unique field (up to isom.)
of order p^d , $\forall$ prime p &
all $d \geq 1$

Notation:

$\mathbb{F}_{p^d}$: finite field
of order p^d